4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Manage Internship Application Record	This is what our application is meant to do at the most basic level.	Inputting required fields, detecting duplicate applications, linking apps to logged in users.
UC-02	Update Application Status and Track History	Application statuses change, and the user must be able to reflect that.	Check if status is one of allowed statuses, not allowing same status changes (applied cannot change to applied)
UC-05	Register and Authenticate Account	Application information must be unique to each user logged in.	Check if information isn't taken, hash password, keep user logged into session

4.2 Expanded use cases

Copy the block below once for each use case in 4.1.

UC-01: Manage Internship Application Record

Primary actor	Student
Supporting actors and external systems	None (as we have no external systems)
TUCBW (this use case begins when)	The use case begins when a student chooses to add a new application or edit/delete an existing one.
TUCEW (this use case ends when)	This use case ends when the student sees requested changes reflected in the application.

Preconditions	The student must have an account and must be logged in. For editing and deleting, the user must have already created an application entry.
Postconditions	If successful, a record is created or saved and reflected in the application, and if “delete” then the application is deleted. If not successful, records are unchanged.
Traces to requirements	FR-03, FR-04, FR-05, FR-06, FR-08, FR-12, QR-02, QR-04, no CON

Actor input / action	System response
	(0) system displays applications list with “add application” option
(1) student selects “Add Application”	(2) system asks for: company name, position title, date applied, deadline, dates for interview, resume and cover letter upload.
(3) Student enters: company name, position title, date applied. Also enters: deadline, interview dates, resume, cover letter, then submits application.	(4) System checks files are PDF less than or equal to 10MB space, and if required fields are filled.
	(5) System reflects changes in application
	(6) System displays application new list
(7) Student sees the new application in their list.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	1	student wants to edit application	student chooses to edit an application’s information and system displays details to change	Rejoins at 3, changes reflected in 7
A2	1	student wants to delete application	The student chooses to delete an application after choosing it. The student must then confirm if they want to delete. If so, delete, if not, do not.	Ends

UC-02: Update Application Status and Track History

Primary actor	Student
Supporting actors and external systems	None
TUCBW (this use case begins when)	The use case begins when the student chooses an existing application.

TUCEW (this use case ends when)	The use case ends when the system displays application edit, or if the student cancels, nothing changes.
Preconditions	The student must be logged in and have at least one application logged.
Postconditions	App shows new changes/edits in the record if successful, and no changes if not successful.
Traces to requirements	FR-04, FR-06, FR-07, FR-08, QR-02

Actor input / action	System response
	(0) system displays all student applications
(1) student chooses application	(2) system displays application info (status: applied, rejected, offered, interviewing)
(3) student chooses new status or new info to fill in and saves it	(4) system checks if new info differs from old info
	(5) system updates current status and adds history entry with new status, previous, and date.
	(6) system displays new status and updated history
(7) student now sees both.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	4	Same status selected	System tells student that nothing has changed and adds no history entry.	ends
A2	3	Student chooses Interviewing	The system asks the student for interview dates, then dates are saved if inputted, with a reminder once within 7 days of interview date.	Rejoins at 5

UC-05: Register and Authenticate Account

Primary actor	Student
Supporting actors and external systems	None (as we manage accounts, not UTD SSO)
TUCBW (this use case begins when)	The student opens InternTrack and chooses to make an account/log in.
TUCEW (this use case ends when)	The student sees the updated status and history, or when the student cancels without making changes

Preconditions	The student must not be logged in
Postconditions	The account is then created (if registering) or logged into if the account already exists. If it fails, no session is started or no account is created.
Traces to requirements	FR-01, FR-02, QR-02, QR-03, CON-01

Actor input / action	System response
	(0) System displays login message with an option to register
(1) Student chooses to create account	(2) System asks for email and password fields to fill out
(3) Student fills out both fields	(4) Check if the email is unique, and that the password is at least 8 characters.
	(5) if so, the system creates an account.
	(6) system tells student account created and to log in
(7) The student enters email, password and logs in.	(8) Check if email and password combinations match an account.
	(9) system now displays the student's applications
(10) student now sees their applications	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	1	Student already has account	The student skips account creation and goes straight to login	Rejoins at 7
A2	8	Email and password combination don't link to an account	System does not grant access to the student and asks them to try again	Rejoins at 7
A3	4	Email is already registered	System tells the student the email is already in use and asks for another email	Rejoins at 3
A4	4	Password does not meet requirements	System displays an error message and asks the student to enter a valid password	Rejoins at 3

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-01/UC-02	Is the actor a role?	Put in “student/registered user” but registered user is not a role.	Changed to just “student”
All three	One turn per row?	Final system row had many functions at once like checking, displaying, recording	Multiple rows for each
UC-02	No background processing?	Included reminder updates student won’t see	Removed as they belong to UC-03
UC-05	Are error cases included?	Missing steps for an email already in use or invalid password	Added A3 and A4 to show what happens